

Math 3IA3 - All Practice Problems

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Section 1 - Practice Problems

Chapter 1.1: Sets and Functions

Exercise 1.1.5 Prove the following distributive laws:

(a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution:

We can prove that two sets are equal by showing that they are both subsets of one another.

(I) Show that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Let $x \in (A \cap (B \cup C))$. Therefore, by definition of intersection and union, we know $x \in A \wedge x \in (B \cup C)$. Thus, this implies that $x \in B \vee x \in C$. However, $x \in A \implies x \in ((A \cap B) \cup (A \cap C)) \iff x \in (B \cup C)$. However, we already determined that $x \in B \vee x \in C$. Thus, $x \in (A \cap (B \cup C)) \implies x \in ((A \cap B) \cup (A \cap C))$ and therefore $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

(II) Show that $A \cap (B \cup C) \supseteq (A \cap B) \cup (A \cap C)$.

Let $x \in ((A \cap B) \cup (A \cap C))$. Therefore, by definition of intersection and union, we know $(x \in A \wedge x \in B) \wedge (x \in A \wedge x \in C)$. This clearly implies that $x \in A$, and furthermore that $x \in C \vee x \in B$. Therefore, we know $x \in (B \cup C)$, and since $x \in A$, $x \in A \cap (B \cup C)$. Thus, $x \in ((A \cap B) \cup (A \cap C)) \implies x \in (A \cap (B \cup C))$ and thus $A \cap (B \cup C) \supseteq (A \cap B) \cup (A \cap C)$.

Therefore, since $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and $A \cap (B \cup C) \supseteq (A \cap B) \cup (A \cap C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Solution:

We will prove this in much the same way as in (a).

(III) Show that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

Let $x \in (A \cup (B \cap C))$. Therefore, $x \in A \vee x \in (B \cap C)$. If $x \in (B \cap C)$ which means we know that $x \in B \wedge x \in C$. Therefore, we know that $x \in A \vee (x \in B \wedge x \in C)$. If $x \in A$, then $x \in (A \cup B) \wedge x \in (A \cup C)$, and therefore $x \in ((A \cup B) \cap (A \cup C))$, and if $x \in B \wedge x \in C$, then $x \in (A \cup B) \wedge x \in (A \cup C)$ and therefore $x \in ((A \cup B) \cap (A \cup C))$. Therefore, $x \in (A \cup (B \cap C)) \implies x \in ((A \cup B) \cap (A \cup C))$ and thus $A \cup (B \cap C) \subseteq ((A \cup B) \cap (A \cup C))$.

(IV) Show that $A \cup (B \cap C) \supseteq (A \cup B) \cap (A \cup C)$.

Let $x \in ((A \cup B) \cap (A \cup C))$. Therefore, $x \in (A \cup B) \wedge x \in (A \cup C)$ which means we know that $x \in A \vee (x \in B \wedge x \in C)$. If $x \in A$, then trivially $x \in (A \cup (B \cap C))$, and if $x \in B \wedge x \in C$, then $x \in (B \cap C)$ and therefore $x \in (A \cup (B \cap C))$. Therefore $x \in ((A \cup B) \cap (A \cup C)) \implies x \in (A \cup (B \cap C))$ and thus $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

Therefore, since $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$ and $A \cup (B \cap C) \supseteq (A \cup B) \cap (A \cup C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Exercise 1.1.11 Let $g(x) := x^2$ and $f(x) := x + 2$ for $x \in \mathbb{R}$, and let h be the composite function $h := g \circ f$.

(a) Find the direct image $h(E)$ of $E := \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

Solution:

Recall that the image of a function under a set is the set of all values which the function can attain with inputs taken from the set. We know the composite function h is given by $h(x) = f(x)^2 = (x + 2)^2$. Therefore, the image of h under E is $\{y \in \mathbb{R} : 4 \leq y \leq 9\}$.

(b) Find the inverse image $h^{-1}(G)$ of $G := \{x \in \mathbb{R} : 0 \leq x \leq 4\}$.

Solution:

Recall that the inverse image of a function is the set of all values in the domain of the function which are in the given set. Since we previously determined that $h(x) = (x + 2)^2$, by rearranging, we can find that:

$$\begin{aligned}x &= (h^{-1}(x) + 2)^2 \\ \pm\sqrt{x} &= (h^{-1}(x) + 2) \\ -2 \pm \sqrt{x} &= h^{-1}(x)\end{aligned}$$

Therefore, $h^{-1}(x) = -2 \pm \sqrt{x}$. Thus, the inverse image of h under G is $\{x \in \mathbb{R} : -2 \leq x \leq 2\}$.

Exercise 1.1.19

- (a) Show that if $f : A \rightarrow B$ is injective and $E \subseteq A$, then $f^{-1}(f(E)) = E$. Give an example to show that equality need not hold if f is not injective.

Solution:

Proof. Let $x \in E$. By definition of inverse, we know trivially that $x \in E \implies x \in f^{-1}(f(E))$ and thus $E \subseteq f^{-1}(f(E))$. Now consider $x \in f^{-1}(f(E))$. To prove that x must necessarily be in E , we rely on a proof by contradiction. Assume for the purposes of contradiction that $x \notin f^{-1}(f(E))$. Since we are taking the inverse of $f(E)$, that implies $\exists y \in E$ such that $f(y) = f(x)$. But since $x \notin E$ and $y \in E$, we necessarily know that $x \neq y$. However, this would violate the assumption that f is injective since, if this were true $\exists x, y \in A, x \neq y$ such that $f(x) = f(y)$. Thus, by contradiction, $x \in f^{-1}(f(E)) \implies x \in E$ and $f^{-1}(f(E)) \subseteq E$.

Therefore, since $E \subseteq f^{-1}(f(E))$ and $f^{-1}(f(E)) \subseteq E$, $E = f^{-1}(f(E))$ for injective f . $\square$

If injectivity is not assumed, this is not necessarily true. Taking $f(x) = x^2$ and $E = [0, 1]$ as an example, $f^{-1}(f(E)) = [-1, 1]$. Clearly then, for non-injective f , equality need not hold.

- (b) Show that if $f : A \rightarrow B$ is surjective and $H \subseteq B$, then $f(f^{-1}(H)) = H$. Give an example to show that equality need not hold if f is not surjective.

Solution:

Proof. Let $K = f^{-1}(H)$. Since f is surjective, we know that $f(K) = H$ and since $K = f^{-1}(H)$, then $H = f(f^{-1}(H))$. Thus, if $x \in f(f^{-1}(H))$, then $x \in H$ and therefore $H \subseteq f(f^{-1}(H))$.

□

Kind of a gross counter example but i'm tired sorry.

If surjectivity is not assumed, this is not necessarily true. Once again, take $f(x) = x^2$ and $H = (-\infty, \infty)$. $f(f^{-1}(H)) = [0, \infty)$. Clearly then, for non-surjective f , equality need not hold.

Exercise 1.1.21 Prove that if $f : A \rightarrow B$ and $g : B \rightarrow C$ are bijective, then the composite $g \circ f$ is a bijective map of A onto C .

Solution:

Proof. To prove that $g \circ f$ is a bijective map of A onto C , we must show that $g \circ f$ is both injective and surjective.

(I) Prove $g \circ f$ is injective.

Since it is given that f and g are injective, then we know $\forall i, j \in A, i \neq j, f(i) \neq f(j)$ and $\forall k, l \in B, g(k) \neq g(l)$. Thus:

$$\begin{aligned} \forall x, y \in A, x \neq y &\implies \\ f(x) \neq f(y) &\implies \\ g(f(x)) \neq g(f(y)) & \end{aligned}$$

Therefore, $g \circ f$ is an injective map of A onto C .

(II) Prove $g \circ f$ is surjective.

Since it is given that f and g are surjective, $i \in B \implies \exists j \in A$ such that $f(j) = i$ and $k \in C \implies \exists l \in B$ such that $g(l) = k$. Thus:

$$\begin{aligned}x \in C &\implies \\ \exists y \in B \text{ such that } g(y) = x &\implies \\ \exists z \in A \text{ such that } f(z) = y &\implies \\ g(f(z)) = x &\end{aligned}$$

Therefore $g \circ f$ is a surjective map of A onto C .

Thus, since $g \circ f$ is both an injective and surjective map of A onto C , $g \circ f$ is bijective.

□

Chapter 1.2: Mathematical Induction

Exercise 1.2.1 Prove that $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n \cdot (n+1)} = \frac{n}{n+1} \quad \forall n \in \mathbb{N}$.

Solution:

Proof. Let $P(n)$ be the statement $P(n) : \frac{1}{1} \cdot 2 + \frac{1}{2} \cdot 3 + \dots + \frac{1}{n \cdot (n+1)} = \frac{n}{n+1}$.

(I) Show the base case holds (Prove $P(1)$ is true).

LHS:

$$\begin{aligned} & \frac{1}{n \cdot (n+1)} \\ &= \frac{1}{1 \cdot (1+1)} \\ &= \frac{1}{2} \end{aligned}$$

RHS:

$$\begin{aligned} & \frac{n}{n+1} \\ &= \frac{1}{1+1} \\ &= \frac{1}{2} \end{aligned}$$

Thus, $P(1)$ is true.

(II) State induction hypothesis (State $P(k)$).

Assuming there exists a $k \in \mathbb{N}$ for which $P(k)$ is true we can formulate our induction hypothesis as:

$$P(k) : \frac{1}{1} \cdot 2 + \frac{1}{2} \cdot 3 + \dots + \frac{1}{k \cdot (k+1)} = \frac{k}{k+1}$$

(III) Prove $P(k) \implies P(k+1)$.

$$\begin{aligned}
 & \frac{1}{1} \cdot 2 + \frac{1}{2} \cdot 3 + \dots + \frac{1}{k \cdot (k+1)} + \frac{1}{(k+1) \cdot (k+2)} \\
 &= \frac{k}{k+1} + \frac{1}{(k+1) \cdot (k+2)} \\
 &= \frac{k(k+2) + 1}{(k+1) \cdot (k+2)} \\
 &= \frac{k^2 + 2k + 1}{(k+1) \cdot (k+2)} \\
 &= \frac{(k+1)^2}{(k+1) \cdot (k+2)} \\
 &= \frac{k+1}{k+2}
 \end{aligned}$$

Therefore, since $P(1)$ is true, and $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$, $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

Exercise 1.2.7 Prove that $5^{2n} - 1$ is divisible by 8 $\forall n \in \mathbb{N}$.

Solution:

Proof. Let $P(n)$ be the statement $P(n) : 8 \mid (5^{2n} - 1)$.

(I) Show the base case holds (Prove $P(1)$ true).

$$\begin{aligned}
 8 \bmod (5^{2^n} - 1) &= \bmod(5^{2 \cdot 1} - 1) \\
 &= 8 \bmod (5^2 - 1) \\
 &= 8 \bmod (24) \\
 &= 0
 \end{aligned}$$

Therefore, $P(1)$ is true.

(II) State induction hypothesis (State $P(k)$).

Assuming there exists a $k \in \mathbb{N}$ for which $P(k)$ is true we can formulate our induction hypothesis as:

$$P(k) : 8 \bmod (5^{2^k} - 1)$$

(III) Prove $P(k) \implies P(k+1)$.

$$\begin{aligned}
 8 \bmod (5^{2^{(k+1)}} - 1) &= 8 \bmod (5^{2^{k+2}} - 1) \\
 &= 8 \bmod (5^2 \cdot 5^{2^k} - 1) \\
 &= 0
 \end{aligned}$$

Therefore, since $P(1)$ is true, and $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$, $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

Exercise 1.2.11 Conjecture a formula for the sum of the first n odd natural numbers $1 + 3 + \dots + (2n - 1)$, and prove your formula using mathematical induction

Solution:

First n odd natural numbers	1	2	3	4	5	...
$1 + 3 + \dots + (2n - 1)$	1	4	9	16	25	...

Based on this, I conjecture that $1 + 3 + \dots + (2n - 1) = n^2$. Next, I will prove this fact.

Proof. Let $P(n)$ be the statement $P(n) : 1 + 3 + \dots + (2n - 1) = n^2$.

(I) Show base case holds (Prove $P(1)$ true).

LHS:	RHS:
$(2n - 1)$	n^2
$= (2(1) - 1)$	$(1)^2$
$= 1$	$= 1$

Therefore, $P(1)$ is true.

(II) State induction hypothesis (State $P(k)$).

Assuming there exists a $k \in \mathbb{N}$ for which $P(k)$ is true we can formulate our induction hypothesis as:

$$P(k) : 1 + 3 + \dots + (2k - 1) = k^2$$

(III) Prove $P(k) \implies P(k + 1)$

$$\begin{aligned} 1 + 3 + \dots + (2k - 1) + (2k + 1) &= k^2 + (2k + 1) \\ &= (k + 1)^2 \end{aligned}$$

Therefore, since $P(1)$ is true, and $P(k) \implies P(k + 1)$ for arbitrary $k \in \mathbb{N}$, $P(n)$ is true $\forall n \in \mathbb{N}$. □

Exercise 1.2.13 Prove that $n < 2^n \forall n \in \mathbb{N}$.

Solution:

Let $P(n)$ be the statement $P(n) : n < 2^n$.

Proof.

(I) Show base case holds (Prove $P(1)$ true).

LHS:

$$n$$

$$= (1)$$

$$= 1$$

RHS:

$$2^n$$

$$= 2^1$$

$$= 2$$

Therefore, $P(1)$ is true.

(II) Formulate induction hypothesis (State $P(k)$).

Assuming there exists a $k \in \mathbb{N}$ for which $P(k)$ is true we can formulate our induction hypothesis as:

$$P(k) : k < 2^k$$

(III) Prove $P(k) \implies P(k+1)$

$$\begin{aligned} k+1 &< 2^k + 1 \\ &< 2^{k+1} \end{aligned}$$

Therefore, since $P(1)$ is true, and $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$, $P(n)$ is true $\forall n \in \mathbb{N}$. $\square$

Chapter 1.3: Finite and Infinite Sets

Exercise 1.3.3 Let $S := \{1, 2\}$ and $T := \{a, b, c\}$

(a) Determine the number of different injections from S into T .

Solution:

My answer is 6 and my reasoning is as follows:

Taking $s_1 \in S$, we have three possible choices of $t_1 \in T$ for which we can create an injection from s_1 to T , giving us 3 different injections. Now taking $s_2 \in S$, we would have three possible choices of $t_2 \in T$, but since the relation must be injective, t_1 cannot equal t_2 and thus we only have two possible choices. Thus, we have $3 \cdot 2 = 6$ possible injections from S to T .

(b) Determine the number of different surjections from T onto S .

Solution:

My answer is 6 and my reasoning is as follows:

Taking $t_1 \in T$, t_1 has two possible choices of $s_1 \in S$ for which we can create a surjection from t_1 to S . Now consider $t_2 \in T$. Since the relationship need not be injective, we separate our choices into two possible cases. If the surjection from t_2 to S is s_1 , then we know that t_3 MUST be s_2 . However, if the surjection from t_2 to S is not s_1 , then t_3 can relate to either of s_1 or s_2 . Thus, we have $2 \cdot 1 \cdot 1 + 2 \cdot 1 \cdot 2 = 6$ possible surjections from T to S .

Exercise 1.3.5 Give an explicit definition of the bijection f from $\mathbb{N}$ onto $\mathbb{Z}$ described in example 1.3.7.(b).

Solution:

$$f(n) = \begin{cases} 0, & \text{if } n = 1 \\ \frac{n}{2}, & \text{if } n \text{ is even} \\ -\frac{n-1}{2}, & \text{if } n \text{ is odd and } n \neq 1 \end{cases}$$

Exercise 1.3.7 Prove that a set T_1 is denumerable if and only if there is a bijection from T_1 onto a denumerable set T_2 .

Solution:

Proof. We prove this by first independently proving the forward and reverse directions of this statement.

(I) Prove the forward direction (T_1 is denumerable $\implies$ there exists a bijection from T_1 to a denumerable set T_2).

Since both T_1 and T_2 are denumerable, we know that there exists two bijection functions f, g such that $f : \mathbb{N} \rightarrow T_1$ and $g : \mathbb{N} \rightarrow T_2$ respectively. However, we know that g^{-1} is also a bijective function from T_2 onto $\mathbb{N}$, and from exercise 1.1.21, we know that the composite function $g^{-1} \circ f$ is a bijection from T_1 onto T_2 . Thus, if T_1 and T_2 are both denumerable, there must exist a bijection from T_1 to T_2 .

(II) Prove the reverse direction (A bijection from T_1 to a denumerable set $T_2 \implies T_1$ is denumerable),

We know that there exists a bijection from T_1 to T_2 . Let's call such a bijection, $f : T_1 \rightarrow T_2$. Since T_2 is denumerable, we know that there exists a bijection from T_2 onto $\mathbb{N}$. Let's call that bijection $g : T_2 \rightarrow \mathbb{N}$. However, from exercise 1.1.21, we proved that $g \circ f$ would be a bijective map from T_1 onto $\mathbb{N}$, making $(f \circ g)^{-1}$ a bijection from $\mathbb{N}$ onto T_1 . Thus, by definition of denumerability, T_1 is denumerable.

Thus, since both the forward and reverse directions are true, T_1 is denumerable $\iff$ there exists a bijection from T_1 to a denumerable set T_2 . $\square$

Exercise 1.3.13 Prove that the collection $\mathcal{F}(\mathbb{N})$ of all *finite* subsets of $\mathbb{N}$ is countable.

Solution:

Proof. To prove that $\mathcal{F}$ is bijective, consider the fact that every natural number can be decomposed into the product of primes. Using this fact, we can construct a bijection between $\mathbb{N}$ and $\mathcal{F}$ with the function:

$$f(n) = \begin{cases} \{1\} & \text{if } n = 1 \\ \{i + 1\} & \text{if } n \text{ is the } (i)^{th} \text{ prime number.} \\ \{j, k\} & \text{if } n \text{ has two prime factors } n_1, n_2 \text{ and where } n_1, n_2 \text{ are the } j^{th} \text{ and } k^{th} \text{ prime numbers respectively} \\ \vdots & \end{cases}$$

□

Section 2 - Practice Problems

Chapter 2.1: The Algebraic and Order Properties of $\mathbb{R}$

Exercise 2.1.1 Solve the following equations, justifying each step by referring to an appropriate property or theorem.

(a) $2x + 5 = 8$.

Solution:

$$2x + 5 = 8 \implies$$

$$2x + 5 - 5 = 8 - 5 \implies \quad \langle \text{By the additive property of equality} \rangle$$

$$2x + 0 = 3 \implies \quad \langle \text{By the additive inverse property} \rangle$$

$$2x = \implies \quad \langle \text{By the additive identity property} \rangle$$

$$\frac{2x}{2} = \frac{3}{2} \implies \quad \langle \text{By the multiplicative property of equality} \rangle$$

$$\frac{2}{2} \cdot x = \frac{3}{2} \implies \quad \langle \text{By the associativity of multiplication} \rangle$$

$$1 \cdot x = \frac{3}{2} \implies \quad \langle \text{By the multiplicative inverse property} \rangle$$

$$x = \frac{3}{2} \quad \langle \text{By the multiplicative identity property} \rangle$$

(b) $x^2 = 2x$

Solution:

$$x^2 = 2x \implies$$

$$x^2 - 2x = 2x - 2x \implies \langle \text{By the additive property of equality} \rangle$$

$$x^2 - 2x = 0 \implies \langle \text{By the additive inverse property} \rangle$$

$$x \cdot (x - 2) = 0 \implies \langle \text{By the distributive property} \rangle$$

$$x = 0 \text{ or } x = 2 \quad \langle \text{By Theorem 2.1.3.b} \rangle$$

(c) $x^2 - 1 = 3$

Solution:

$$x^2 - 1 = 3 \implies$$

$$x^2 - 1 - 3 = 3 - 3 \implies \langle \text{By the additive property of equality} \rangle$$

$$x^2 - 4 = 0 \implies \langle \text{By the additive inverse property} \rangle$$

$$(x + 2) \cdot (x - 2) = 0 \implies \langle \text{By the distributive property} \rangle$$

$$x = 2 \vee x = -2 \quad \langle \text{By Theorem 2.1.3.a} \rangle$$

(d) $(x - 1)(x - 2) = 0$

Solution:

$$(x - 1)(x - 2) = 0 \implies$$

$$x = 1 \vee x = 2 \quad \langle \text{By Theorem 2.1.3.a} \rangle$$

Exercise 2.1.5 If $a \neq 0$ and $b \neq 0$, show that $\frac{1}{ab} = (\frac{1}{a})(\frac{1}{b})$.

Solution:

$$\begin{aligned} \frac{1}{ab} &= 1 \div ab \\ &= 1 \cdot \frac{1}{ab} \\ &= (1 + 0 + 0) \cdot \frac{1}{ab} \\ &= (1 + a - a + b - b) \cdot \frac{1}{ab} \\ &= (1 \cdot \frac{1}{ab} + \frac{a}{ab} - \frac{a}{ab} + \frac{b}{ab} - \frac{b}{ab}) \end{aligned}$$

Exercise 2.1.9 Let $K := \{s + t\sqrt{2} : s, t \in \mathbb{Q}\}$. Show that K satisfies the following:

(a) If $x_1, x_2 \in K$, then $x_1 + x_2 \in K$ and $x_1x_2 \in K$.

Solution:

Let $x_1 = s_1 + t_1\sqrt{2}, x_2 = s_2 + t_2\sqrt{2} \in K$. We separate the proof of each claim into two parts into two parts.

(I) Prove $x_1 + x_2 \in K$

Proof. We can see that:

$$\begin{aligned}x_1 + x_2 &= s_1 + t_1\sqrt{2} + s_2 + t_2\sqrt{2} \\ &= (s_1 + s_2) + (t_1 + t_2)\sqrt{2}\end{aligned}$$

Thus, since $\mathbb{Q}$ is closed under addition, we know that $(s_1 + s_2) \in \mathbb{Q}$ and $(t_1 + t_2) \in \mathbb{Q}$. Therefore, $((s_1 + s_2) + (t_1 + t_2)\sqrt{2}) \in K$ and thus $(x_1 + x_2) \in K$. $\square$

(II) Prove $x_1x_2 \in K$

Proof. We can see that:

$$\begin{aligned}x_1x_2 &= (s_1 + t_1\sqrt{2})(s_2 + t_2\sqrt{2}) \\ &= s_1s_2 + s_1t_2\sqrt{2} + s_2t_1\sqrt{2} + t_1t_2\sqrt{2}\sqrt{2} \\ &= s_1s_2 + (s_1t_2 + s_2t_1)\sqrt{2} + t_1t_2(\sqrt{2})^2 \\ &= s_1s_2 + (s_1t_2 + s_2t_1)\sqrt{2} + 2t_1t_2 \\ &= (s_1s_2 + 2t_1t_2) + (s_1t_2 + s_2t_1)\sqrt{2}\end{aligned}$$

Thus, since $\mathbb{Q}$ is closed under both addition and multiplication, we know that $s_1s_2 + 2t_1t_2 \in \mathbb{Q}$ and $s_1t_2 + s_2t_1 \in \mathbb{Q}$. Therefore, $(s_1s_2 + 2t_1t_2) + (s_1t_2 + s_2t_1)\sqrt{2} \in K$ and thus $x_1x_2 \in K$. $\square$

(b) If $x \neq 0$ and $x \in K$, then $\frac{1}{x} \in K$.

Solution:

Proof. Let $x = s + t\sqrt{2} \in K$, where s and t cannot equal 0 simultaneously. Therefore, we have:

$$\begin{aligned}
\frac{1}{x} &= \frac{1}{s + t\sqrt{2}} \\
&= \frac{1}{s + t\sqrt{2}} \cdot \frac{s - t\sqrt{2}}{s - t\sqrt{2}} \\
&= \frac{s - t\sqrt{2}}{(s + t\sqrt{2})(s - t\sqrt{2})} \\
&= \frac{s - t\sqrt{2}}{s^2 + (t\sqrt{2})^2} \\
&= \frac{s - t\sqrt{2}}{s^2 + 2t^2} \\
&= \frac{s}{s^2 + 2t^2} - \frac{t}{s^2 + 2t^2} \sqrt{2}
\end{aligned}$$

Therefore, since $\mathbb{Q}$ is closed under addition, multiplication, and division, we know $\frac{s}{s^2+2t^2} \in \mathbb{Q}$ and $\frac{t}{s^2+2t^2} \in \mathbb{Q}$. Therefore, $\frac{s}{s^2+2t^2} - \frac{t}{s^2+2t^2} \sqrt{2} \in \mathbb{Q}$ and thus $\frac{1}{x} \in K$ if $x \neq 0$. $\square$

Exercise 2.1.15 If $0 < a < b$, prove that:

(a) $a < \sqrt{ab} < b$

Solution:

Proof.

$$a < b \implies$$

$$\begin{aligned}
ab < b \cdot b &\implies a \cdot a < ab && \implies \\
ab < b^2 &\implies a^2 < ab && \implies \\
\sqrt{ab} < \sqrt{b^2} &\implies \sqrt{a^2} < \sqrt{ab} && \implies \\
\sqrt{ab} < b &a < \sqrt{ab}
\end{aligned}$$

Therefore, $0 < a < b \implies a < \sqrt{ab} < b$. $\square$

(b) $\frac{1}{b} < \frac{1}{a}$

Solution:

Proof.

$$\begin{aligned} a &< b \\ \frac{a}{1} &< \frac{b}{1} \\ \frac{a}{1} \cdot 1 &< \frac{b}{1} \cdot 1 \\ \frac{a}{1} \cdot \frac{b}{b} &< \frac{b}{1} \cdot \frac{a}{a} \\ \frac{ab}{b} &< \frac{ab}{a} \\ \frac{ab}{b} \cdot \frac{1}{ab} &< \frac{ab}{a} \cdot \frac{1}{ab} \\ \frac{1}{b} &< \frac{1}{a} \end{aligned}$$

Therefore, $0 < a < b \implies \frac{1}{b} < \frac{1}{a}$.

□

Chapter 2.2: Absolute Value and the Real Line

Exercise 2.2.9 Find all values of x that satisfy the following inequalities. Sketch all graphs.

(a) $|x - 2| \leq x + 1$

Solution:

$$\begin{array}{ll} |x - 2| \leq x + 1 & \implies \\ x - 2 \leq x + 1 & x - 2 \leq -(x + 1) \\ & x - 2 \leq -x - 1 \\ & 2x \leq 1 \\ & x \leq \frac{1}{2} \end{array}$$

Therefore, the inequality holds whenever $x \in (-\infty, \frac{1}{2}]$.

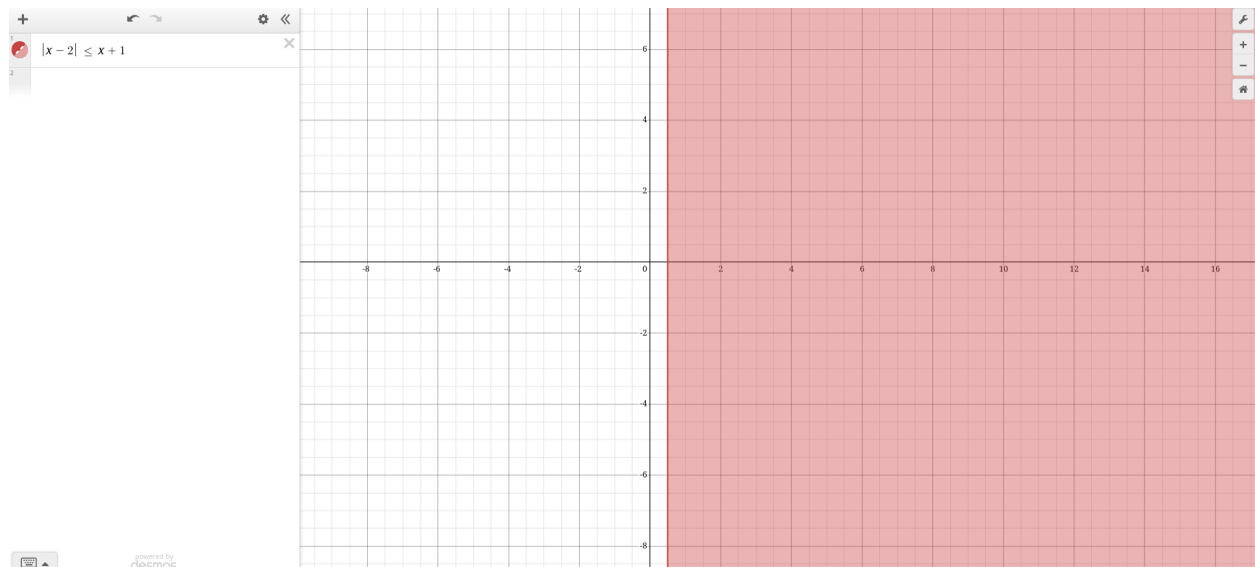


Figure 1: $|x - 2| \leq x + 1$ as graphed by Desmos.

(b) $3|x| \leq 2 - x$

Solution:

$$\begin{array}{llll}
 3|x| & \leq 2 - x & \implies \\
 3(x) \leq 2 - x & 3x \leq -(2 - x) & \implies \\
 4x \leq 2 & 3x \leq 2 + x & \implies \\
 x \leq \frac{1}{2} & 4x \geq 2 & \\
 \end{array}$$

Therefore, the inequality holds whenever $x \in (-\infty, \frac{1}{2}] \cup [1, \infty)$.

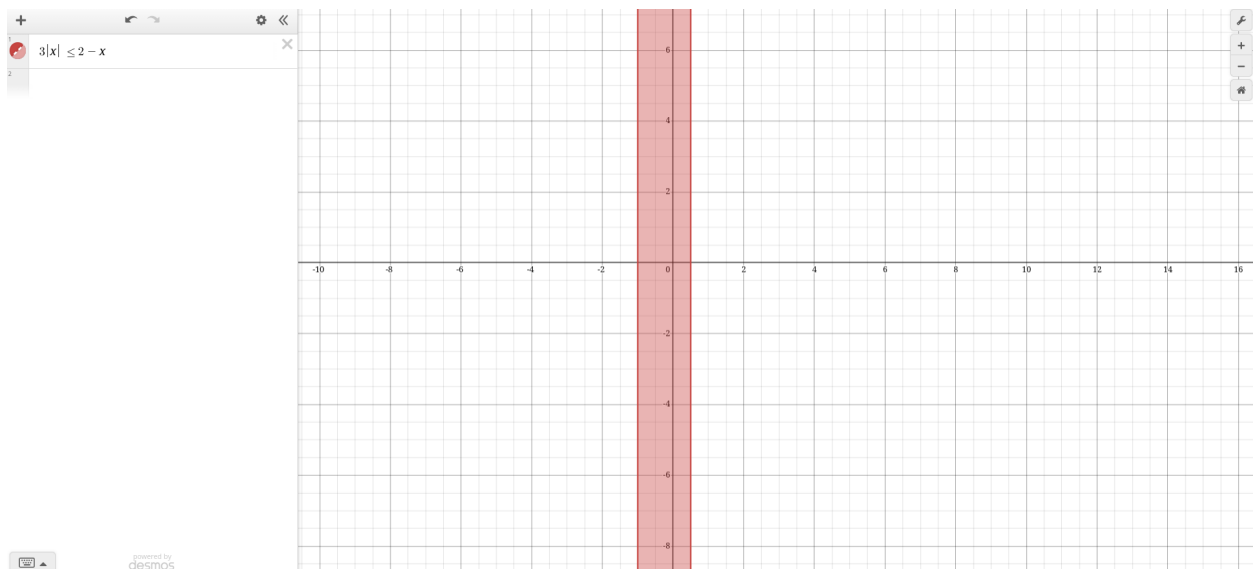


Figure 2: $3|x| \leq 2 - x$ as graphed by Desmos.

Exercise 2.2.14 Determine and sketch the set of pairs $(x, y) \in \mathbb{R} \times \mathbb{R}$ that satisfy:

(a) $|x| = |y|$.

Solution:

$$\begin{aligned} |x| = |y| &\implies \\ (x = y \vee x = -y \vee -x = y \vee -x = -y) &\implies \\ x = y \vee x = -y & \end{aligned}$$

(b) $|x| + |y| = 1$.

(c) $|xy| = 2$.

(d) $|x| - |y| = 2$.

Exercise 2.2.19 Show that if $a, b, c \in \mathbb{R}$, then the “middle number” is $\text{mid}\{a, b, c\} = \min\{\max\{a, b\}, \max\{b, c\}, \max\{c, a\}\}$.

Solution:

Chapter 2.3: The Completeness Property of $\mathbb{R}$

Exercise 2.3.1 Let $S_3 = \{\frac{1}{n} : n \in \mathbb{N}\}$. Show that $\sup S_3 = 1$ and $\inf S_3 \geq 0$. (It will follow from the Archimedean Property in Section 2.4 that $\inf S_3 = 0$.)

Solution:

(I) Show that $\sup S_3 = 1$

By definition of supremum, we know that 1 is the supremum of S_3 if, for all $\epsilon > 0$:

$$1 \geq s_3 \quad \forall s_3 \in S_3 \text{ AND } \exists s_3 \in S_3 \text{ such that } s_3 > 1 - \epsilon.$$

Without appealing to the archimedean property, we know that 1 must be the supremum of this set because

(II) Show that $\inf S_3 \geq 0$.

Since $n \in \mathbb{N}$, we know $n \geq 0$ by the order limit properties of $\mathbb{R}$ that $\frac{1}{n} \geq 0$ for all $n \in \mathbb{N}$. Now, let's assume for the purposes of contradiction that $s_1 = \inf S_3 < 0$. In such a case, we know that there exists an $s_2 \in \mathbb{R}$ such that $s_1 > s_2 > 0$. However, we know that s_2 must also lower bound S_3 since $\frac{1}{n} > 0 \quad \forall n \in \mathbb{N}$. Thus, any $s_1 > 0$ cannot be the infimum of S_3 since there will always exist a number larger than it (s_2) which lower bounds S_3 . Therefore, by contradiction, the infimum of S_3 must be greater than or equal to zero.

Exercise 2.3.6 Let S be a nonempty subset of $\mathbb{R}$ that is bounded below. Prove that $\inf(S) = -\sup(\{-s : s \in S\})$.

Solution:

Proof. We can appeal to the completeness property of $\mathbb{R}$ to assert that $\inf(S)$ must exist since S is bounded below. Let $m = \inf(S)$. By definition of infimum, we know that for all $s \in S$, $m \leq s$. Therefore, we know that $-m \geq -s$. Therefore, for the set $K = \{-s : s \in S\}$, $-m$ upper bounds K . Now we must prove that $-m$ is the supremum of K . Since m is the infimum of S , we know that $\forall \epsilon > 0, \exists s \in S$ such that $m + \epsilon > s$. Therefore, we have $-m - \epsilon < -s$. Therefore, since m upper bounds K , and for arbitrary $\epsilon > 0$, there always

exists a $k \in K$ such that $-m - \epsilon < k$, $-m$ must be the supremum of $K = \{-s : s \in S\}$. Therefore, appealing to our earlier definition of m , we have:

$$\begin{aligned} m &= m \\ m &= -(-m) \\ \inf(S) &= -\sup(K) \\ \inf(S) &= -\sup(\{-s : s \in S\}) \end{aligned}$$

□

Exercise 2.3.10 Show that if A and B are bounded subsets of $\mathbb{R}$, then $A \cup B$ is a bounded set. Show that $\sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$.

Solution:

We separate our proof into two parts.

(I) Prove that $A \cup B$ is bounded.

Proof. By definition of boundedness, we know that $\exists M_1, M_2 \in \mathbb{R}$ such that $|a| \leq M_1, |b| \leq M_2 \forall a, b \in A, B$. Let $M = \max M_1, M_2$. Thus, we trivially have $M \geq M_1$ and $M \geq M_2$. We know that $\forall c \in (A \cup B), c \in A \vee c \in B$. If $c \in A$, then $|c| \leq M$ since $M \geq M_1$ and if $c \in B$, $|c| \leq M$ since $M \geq M_2$. Thus, $A \cup B$ must be bounded for bounded A, B . □

(II) Prove that $\sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$.

Proof. Appealing to the completeness property of $\mathbb{R}$, since A and B are both bounded, we know that $\sup(A), \sup(B)$ both exist. Let $s_1 = \sup(A)$ and $s_2 = \sup(B)$, and $\epsilon > 0$ be arbitrary. By definition of suprema, we know that $\exists a_0, b_0 \in A, B$:

$$s_1 - \epsilon < a_0 \quad \text{and} \quad s_2 - \epsilon < b_0$$

Now, let $s_3 = \sup(\{s_1, s_2\})$. Since the set $\{s_1, s_2\}$ is finite, we know that the supremum must be an element of the set and thus either $s_3 = s_1$ or $s_3 = s_2$ and therefore $s_3 \geq s_1$ and $s_3 \geq s_2$. Let $c \in A \cup B$. Since $c \in A \vee c \in B$, we know trivially that $c \leq s_3$ meaning s_3 upperbounds $A \cup B$. To show that s_3 must be the supremum, consider again $\epsilon > 0$. From

the properties of s_1 and s_2 that we discussed earlier, we know that if $s_3 = s_1$, $\exists a_0 \in A$ such that $s_3 - \epsilon < a_0$ and if $s_3 = s_2$, $\exists b_0 \in B$ such that $s_3 - \epsilon < b_0$ for all ϵ . Since $a_0, b_0 \in A \cup B$, there always exists a $c_0 \in A \cup B$ such that:

$$s_3 - \epsilon < c_0$$

Thus, by definition of supremum, we know $s_3 = \sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$. $\square$

Chapter 2.4: Application of the Supremum Property

Exercise 2.4.1 If $S := \{\frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N}\}$, find $\inf(S)$ and $\sup(S)$.

Solution:

I assert that $\inf(S) = -1$ and $\sup(S) = 1$ and my reasoning is as follows:

$$\begin{array}{ccc} 0 < \frac{1}{n} \leq 1, & 0 < \frac{1}{m} \leq 1, & \forall m, n \in \mathbb{N} \implies \\ \frac{1}{n} - \frac{1}{m} \geq 0 - 1 & & \frac{1}{n} - \frac{1}{m} \leq 1 - 0 \\ = -1 & & = 1 \end{array}$$

Therefore, -1 and 1 clearly upper and lowerbound S respectively.

To prove that -1 and 1 are the infimum and supremum of S , consider $\epsilon > 0$. Now consider $1 - \epsilon$. By the Archimedean principle, we know that there exists an $n \in \mathbb{N}$ such that $N > 1 - \epsilon$ and

Chapter 2.5: Intervals

Section 3 - Practice Problems

Chapter 3.1: Sequences and Their Limits

Exercise 3.1.7 Let $x_n := \frac{1}{\ln(n+1)}$ for $n \in \mathbb{N}$.

(a) Use the definition of limits to show that $\lim_{n \rightarrow \infty} (x_n) = 0$.

Solution:

Let $\epsilon > 0$ be arbitrary. Since $\ln$ is an increasing function, we know that $\ln(n) \leq \ln(n+1)$ and thus $\frac{1}{\ln(n)} \geq \frac{1}{\ln(n+1)}$. Let $N > e^{\frac{1}{\epsilon}}$. We therefore know that $\forall n \geq N$:

$$\begin{aligned} |x_n - 0| &= |x_n| \\ &= \left| \frac{1}{\ln(n+1)} \right| \\ &= \frac{1}{\ln(n+1)} \\ &\leq \frac{1}{\ln(n)} \\ &\leq \frac{1}{\ln(N)} \\ &< \frac{1}{\ln(e^{\frac{1}{\epsilon}})} \\ &= \frac{1}{\frac{1}{\epsilon}} \\ &= \epsilon \end{aligned}$$

Therefore, by definition of limits, $\lim_{n \rightarrow \infty} (x_n) = 0$.

(b) Find a specific value of N as required in the definition of limits for each of the following ϵ -challenges:

(I) $\epsilon = \frac{1}{2}$.

Solution:

From our prior definition of N , we can select any natural number greater than:

$$\begin{aligned} e^{\frac{1}{\epsilon}} &= e^{\frac{1}{2}} \\ &= e^2 \\ &\approx 7.3890561 \end{aligned}$$

Thus, we can choose $N = 8$ to satisfy this ϵ -challenge.

(II) $\epsilon = \frac{1}{10}$

Solution:

For the same reason outlined earlier, we must select an N greater than:

$$\begin{aligned} e^{\frac{1}{\epsilon}} &= e^{\frac{1}{10}} \\ &= e^{10} \\ &\approx 22026.46 \end{aligned}$$

Thus, we can choose $N = 22027$ to satisfy this ϵ -challenge.

Exercise 3.1.8

(a) Prove that $\lim_{n \rightarrow \infty} (x_n) = 0$ if and only if $\lim_{n \rightarrow \infty} (|x_n|) = 0$.

Solution:

Proof.

(I) Prove forward direction ($\lim_{n \rightarrow \infty} (x_n) = 0 \implies \lim_{n \rightarrow \infty} (|x_n|) = 0$)

Let $\epsilon > 0$ be arbitrary.

We know that the sequence $(|x_n|)$ converges to zero if for any such ϵ , there exists some $N \in \mathbb{N}$ such that $\forall n \geq N$, $||x_n| - 0| < \epsilon$.

Let's briefly turn our attention to the sequence (x_n) . Since x_n converges to zero, then we know that there exists an $N_0 \in \mathbb{N}$ such that $\forall n \geq N_0$, $|x_n - 0| < \epsilon$. However, considering that $||x_n| = |x_n|$, we necessarily have, $\forall n \geq N_0$:

$$\begin{aligned} ||x_n| - 0| &= ||x_n|| \\ &= |x_n| \\ &= |x_n - 0| \\ &< \epsilon \end{aligned}$$

Thus, since $||x_n| - 0| < \epsilon$ for arbitrary $\epsilon > 0$ so long as $n \geq N_0$, by definition of convergence, the sequence $(|x_n|)$ converges to zero and thus $\lim_{n \rightarrow \infty}(x_n) = 0 \implies \lim_{n \rightarrow \infty}(|x_n|) = 0$.

(II) Prove reverse direction ($\lim_{n \rightarrow \infty}(|x_n|) = 0 \implies \lim_{n \rightarrow \infty}(x_n) = 0$).

Once again, let $\epsilon > 0$ be arbitrary.

We know that the sequence (x_n) converges to zero if, for any such ϵ , there exists an $N \in \mathbb{N}$ such that $\forall n \geq N$, $|x_n - 0| < \epsilon$.

Using the same idea as before, we know that there exists an $N_0 \in \mathbb{N}$ such that $\forall n \geq N$, $||x_n| - 0| < \epsilon$ since $(|x_n|)$ converges to zero. Thus, $\forall n \geq N_0$:

$$\begin{aligned} |x_n - 0| &= |x_n| \\ &= ||x_n|| \\ &= ||x_n| - 0| \\ &< \epsilon \end{aligned}$$

Thus, since $|x_n - 0| < \epsilon$ for arbitrary $\epsilon > 0$ so long as $n \geq N_0$, by definition of convergence, the sequence (x_n) converges to zero and thus $\lim_{n \rightarrow \infty}(|x_n|) = 0 \implies \lim_{n \rightarrow \infty}(x_n) = 0$.

Therefore, since:

$$\lim_{n \rightarrow \infty}(x_n) = 0 \implies \lim_{n \rightarrow \infty}(|x_n|) = 0$$

and

$$\lim_{n \rightarrow \infty}(|x_n|) = 0 \implies \lim_{n \rightarrow \infty}(x_n) = 0$$

$$\lim_{n \rightarrow \infty}(x_n) = 0 \iff \lim_{n \rightarrow \infty}(|x_n|) = 0$$

□

- (b) Give an example to show that the convergence of $|x_n|$ need not imply the convergence of (x_n) .

Solution:

The sequence $x_n = (-1)^n$ is a very simple example which illustrates this fact. Clearly, $x_n = (-1)^n$ is divergent, however $|x_n| = 1 \forall n \in \mathbb{N}$ and thus $|x_n|$ is clearly convergent to 1. Thus, $|x_n|$ being convergent does not necessarily imply that x_n is.

Exercise 3.1.14 Let $b \in \mathbb{R}$ satisfy $0 < b < 1$. Show that $\lim_{n \rightarrow \infty} (nb^n) = 0$.

Solution:

Exercise 3.1.17 Show that $\lim_{n \rightarrow \infty} (\frac{2^n}{n!}) = 0$. [Hint: If $n \geq 3$, then $0 < \frac{2^n}{n!} < 2 \cdot (\frac{2}{3})^{n-2}$]

Solution:

Let $\epsilon > 0$ be arbitrary. To prove that $\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$, we must show that there exists some $N \in \mathbb{N}$ such that $n \geq N \implies |\frac{2^n}{n!} - 0| < \epsilon$. To start, we begin by setting $N_1 = 3$. Thus, using the hint given in the question, $\forall n \geq N_1$, we have:

$$\begin{aligned} \left| \frac{2^n}{n!} - 0 \right| &= \left| \frac{2^n}{n!} \right| \\ &\leq \left| 2 \cdot \left(\frac{2}{3} \right)^{n-2} \right| \\ &= \left| \frac{9}{2} \left(\frac{2}{3} \right)^n \right| \\ &= \frac{9}{2} \left(\frac{2}{3} \right)^n \end{aligned}$$

Now let's consider the sequence $(\frac{2}{3})^n$. Our goal is to prove that this sequence converges to zero, because if it does, then we can make that term arbitrarily small in our previous

expression and our proof that $\lim_{n \rightarrow \infty} (\frac{2^n}{N!}) = 0$ is complete. Let $N_2 > \log_{\frac{2}{3}}(\frac{2\epsilon}{9})$, we know that $\forall n \geq N_2 \implies$:

$$\begin{aligned} |(\frac{2}{3})^n - 0| &= |(\frac{2}{3})^n| \\ &\geq |(\frac{2}{3})^{N_2}| < |(\frac{2}{3})^{\log_{\frac{2}{3}}(\frac{2\epsilon}{9})}| \\ &= |\frac{2\epsilon}{9}| \\ &= \frac{2\epsilon}{9} \end{aligned}$$

Therefore, by definition of convergence, we know that $\lim_{n \rightarrow \infty} (\frac{2}{3})^n = 0$. Thus, by letting $N = \max\{N_1, N_2\}$, then $n \geq N \implies$:

$$\begin{aligned} |\frac{2^n}{n!} - 0| &= |\frac{2^n}{n!}| \\ &\leq |2 \cdot (\frac{2}{3})^{n-2}| \\ &= |\frac{9}{2} \cdot (\frac{2}{3})^n| \\ &= \frac{9}{2} \cdot (\frac{2}{3})^n \\ &< \frac{9}{2} \cdot \frac{2\epsilon}{9} \\ &= \epsilon \end{aligned}$$

Hence, by definition of convergence, $\lim_{n \rightarrow \infty} (\frac{2^n}{n!}) = 0$.

Exercise 3.1.18 If $\lim_{n \rightarrow \infty} (x_n) = x > 0$, show that there exists a $K \in \mathbb{N}$ such that if $n \geq K$, then $\frac{1}{2}x < x_n < 2x$.

Solution:

Let $\epsilon = \frac{1}{2}x$ (which we can do since $x > 0$). By definition of convergence, we know that there exists a $K \in \mathbb{N}$ such that $\forall n \geq K$:

$$|x_n - x| < \epsilon = \frac{1}{2}x$$

However, notice that:

$$\begin{aligned} |x_n - x| &< \frac{1}{2}x \implies \\ -\frac{1}{2}x &< x_n - x < \frac{1}{2}x \implies \\ -\frac{1}{2}x + x &< x_n - x + x < \frac{1}{2}x + x \implies \\ \frac{1}{2}x &< x_n < \frac{3}{2}x \implies \\ \frac{1}{2}x &< x_n < 2x \end{aligned}$$

Thus, $\forall n \geq K, \frac{1}{2}x < x_n < 2x$.

Chapter 3.2: Limit Theorems

Exercise 3.2.7 If (b_n) is a bounded sequence and $\lim_{n \rightarrow \infty} (a_n) = 0$, prove that $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = 0$. Explain why Theorem 3.2.3 *cannot* be used.

Solution:

Proof. Let $\epsilon > 0$ be arbitrary. Since b_n is given to be bounded, we know that there exists an $M \in \mathbb{R}, M > 0$ such that $|b_n| \leq M \ \forall n \in \mathbb{N}$. Furthermore, because a_n converges to 0, we know that there exists an $N \in \mathbb{N}$ such that $n \geq N \implies |a_n - 0| = |a_n| < \frac{\epsilon}{M}$. Thus, $n \geq N \implies$:

$$\begin{aligned} |a_n \cdot b_n - 0| &= |a_n \cdot b_n| \\ &\leq M \cdot |a_n| \\ &< M \cdot \frac{\epsilon}{M} \\ &< \epsilon \end{aligned}$$

Thus, by definition of convergence, $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = 0$. $\square$

We could not have simply invoked Theorem 3.2.3 to prove this since (b_n) is simply given to be bounded, not convergent.

Exercise 3.2.8 Explain why the result in equation (3) before theorem 3.2.4 *cannot* be used to evaluate the limit of the sequence $\lim_{n \rightarrow \infty} ((1 + \frac{1}{n})^n)$.

Solution:

Equation 3 states that:

$$\lim_{n \rightarrow \infty} (a_n^k) = (\lim_{n \rightarrow \infty} (a_n))^k, \quad \forall k \in \mathbb{N}$$

We cannot use this to evaluate the limit of the sequence $\lim_{n \rightarrow \infty} ((1 + \frac{1}{n})^n)$ since the exponent here is not a constant k but rather a limiting variable. If we were to attempt to apply this law, we would get that $\lim_{n \rightarrow \infty} ((1 + \frac{1}{n})^n) = 1$, which as was determined earlier, would be incorrect.

Exercise 3.2.11 Determine the following limits:

(a) $\lim_{n \rightarrow \infty} ((3\sqrt{n})^{\frac{1}{2n}})$.

Exercise 3.2.16 Apply theorem 3.2.11 to the following sequences, where a, b satisfy $0 < a < 1, b > 1$.

(a) (a^n) .

Solution:

Theorem 3.2.11 states that for a sequence (x_n) of real numbers, that if the sequence $(\frac{x_{n+1}}{x_n})$ exists and is less than 1, then (x_n) converges and $\lim_{n \rightarrow \infty} (x_n) = 0$

For the sequence a^n , we know that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{a^{n+1}}{a^n} \right) &= \lim_{n \rightarrow \infty} (a) \\ &= a \end{aligned}$$

And since $0 < a < 1$, we know that the sequence a^n converges and $\lim_{n \rightarrow \infty} a^n = 0$.

(b) $(\frac{b^n}{2^n})$.

Solution:

$$\lim_{n \rightarrow \infty} \left(\frac{b^{n+1}/2^{n+1}}{b^n/2^n} \right) = \lim_{n \rightarrow \infty} \left(\frac{b}{2} \right)$$

Therefore, $\forall b > 2$, the sequence diverges, and for all $1 < b < 2$ the sequence converges to 0. If $b = 2$, we get that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{b^n}{2^n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{2^n}{2^n} \right) \\ &= \lim_{n \rightarrow \infty} (1) \\ &= 1 \end{aligned}$$

Therefore, when $b = 2$, the sequence converges to 1.

(c) $\left(\frac{n}{b^n}\right)$.

Solution:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left(\frac{n + 1/b^{n+1}}{n/b^n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{n + 1}{b^{n+1}} \cdot \frac{b^n}{n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{b^n(n + 1)}{b^{n+1}(n)} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{n + 1}{b(n)} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{b} + \frac{1}{b(n)} \right) \\ &= \frac{1}{b}\end{aligned}$$

Therefore, since $b > 1 \implies \frac{1}{b} < 1$, we know that $\lim_{n \rightarrow \infty} \left(\frac{n+1/b^{n+1}}{n/b^n} \right) < 1$ and thus $\frac{n}{b^n}$ converges to zero.

(d) $\left(\frac{2^{3n}}{3^{2n}}\right)$.

Solution:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left(\frac{2^{3(n+1)}/3^{2(n+1)}}{2^{3n}/3^{2n}} \right) &= \lim_{n \rightarrow \infty} \left(\frac{(2^3 \cdot 2^{3n})/(3^2 \cdot 3^{2n})}{(2^{3n})/(3^{2n})} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2^3}{3^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{8}{9} \right) \\ &= \frac{8}{9}\end{aligned}$$

Therefore, since $\lim_{n \rightarrow \infty} \left(\frac{2^{3(n+1)}/3^{2(n+1)}}{2^{3n}/3^{2n}} \right) = \frac{8}{9} < 1$, the sequence $\left(\frac{2^{3n}}{3^{2n}} \right)$ converges to zero.

Exercise 3.2.17

(a) Give an example of a convergent sequence (x_n) of positive numbers with $\lim_{n \rightarrow \infty} \left(\frac{x_{n+1}}{x_n} \right)$.

Solution:

The sequence $x_n = 1$ is trivially convergent to 1 and has the property that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{x_{n+1}}{x_n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{1}{1} \right) \\ &= \lim_{n \rightarrow \infty} (1) \\ &= 1 \end{aligned}$$

(b) Give an example of a divergent sequence with this property. (Thus, this property cannot be used as a test for convergence).

Solution:

The sequence $x_n = n$ is an example of a sequence which is clearly divergent (since (x_n) is unbounded), with the property that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{x_{n+1}}{x_n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right) \\ &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) \\ &= \lim_{n \rightarrow \infty} (1) + \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right) \\ &= 1 + 0 \\ &= 1 \end{aligned}$$

Exercise 3.2.19 Discuss the convergence of each of the following sequences, where a, b satisfy $0 < a < 1, b > 1$.

(a) $(n^2 a^n)$.

Solution:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left(\frac{(n+1)^2 / a^{n+1}}{n^2 / a^n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2}{a^{n+1}} \cdot \frac{a^n}{n^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2 (a^n)}{a^{n+1} (n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2}{a(n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{n^2 + 2n + 1}{a(n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{a} + \frac{2}{a(n)} + \frac{1}{a(n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{a} \right) + \lim_{n \rightarrow \infty} \left(\frac{2}{a(n)} \right) + \lim_{n \rightarrow \infty} \left(\frac{1}{a(n)^2} \right) \\ &= \frac{1}{a}\end{aligned}$$

Since $0 < a < 1 \implies \frac{1}{a} > 1$, $\lim_{n \rightarrow \infty} \left(\frac{(n+1)^2 / a^{n+1}}{n^2 / a^n} \right) > 1$ and thus the sequence is divergent.

(b) $\frac{b^n}{n^2}$.

Solution:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left(\frac{b^{n+1}/(n+1)^2}{b^n/n^2} \right) &= \lim_{n \rightarrow \infty} \left(\frac{b^{n+1}}{(n+1)^2} \cdot \frac{n^2}{b^n} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{b^{n+1}(n)^2}{(n+1)^2 b^n} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{b(n)^2}{(n+1)^2} \right) \\
&= \lim_{n \rightarrow \infty} (b) \cdot \lim_{n \rightarrow \infty} \left(\frac{n^2}{(n+1)^2} \right) \\
&= b \cdot 1 \\
&= b
\end{aligned}$$

Therefore, since $b > 1$, $\lim_{n \rightarrow \infty} \left(\frac{b^{n+1}/(n+1)^2}{b^n/n^2} \right) > 1$ and thus the sequence is divergent.

(c) $\frac{b^n}{n!}$

Solution:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left(\frac{b^{n+1}/(n+1)!}{b^n/n!} \right) &= \lim_{n \rightarrow \infty} \left(\frac{b^{n+1}}{(n+1)!} \cdot \frac{n!}{b^n} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{b^{n+1}n!}{b^n(n+1)!} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{(b)n!}{(n+1)!} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{(b)n!}{(n+1)n!} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{b}{n+1} \right) \\
&= 0
\end{aligned}$$

Therefore, since $\lim_{n \rightarrow \infty} \frac{b^{n+1}/(n+1)!}{b^n/n!} = 0 < 1$, the sequence is convergent.

(d) $\frac{n!}{n^n}$

Solution:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \left(\frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)!n^n}{n!(n+1)^{n+1}} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)n!n^n}{n!(n+1)^{n+1}} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)n^n}{(n+1)^{n+1}} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{n^n}{n+1^n} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\left(\frac{n}{n+1} \right)^n \right) \\
 &= \lim_{n \rightarrow \infty} \left(\left(1 - \frac{1}{n+1} \right)^n \right) \\
 &< \lim_{n \rightarrow \infty} (1)^n \\
 &= 1
 \end{aligned}$$

Therefore, since $\lim_{n \rightarrow \infty} \left(\frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} \right) < 1$, the sequence is convergent.

Exercise 3.2.20 Let (x_n) be a sequence of positive numbers such that $\lim_{n \rightarrow \infty} ((x_n)^{\frac{1}{n}}) = L < 1$. Show that there exists a number r with $0 < r < 1$ such that $0 < x_n < r^n$ for all sufficiently large $n \in \mathbb{N}$. Use this to show that $\lim_{n \rightarrow \infty} (x_n) = 0$.

Solution:

Let $\epsilon \in (0, 1 - L)$ and let $r = L + \epsilon$ (clearly then $0 < r < 1$). By definition of convergence, since the sequence $\lim_{n \rightarrow \infty} ((x_n)^{\frac{1}{n}})$ converges to L , we know that there exists an $N \in \mathbb{N}$ such that $\forall n \geq N$:

$$|x_n^{\frac{1}{n}} - L| < \epsilon$$

But that means:

$$\begin{aligned}
(x_n)^{\frac{1}{n}} - L < \epsilon &\implies \\
(x_n)^{\frac{1}{n}} < \epsilon + L &\implies \\
((x_n)^{\frac{1}{n}})^n < (\epsilon + L)^n &\implies \\
x_n < (\epsilon + L)^n &\implies \\
x_n < r^n
\end{aligned}$$

Therefore, since x_n is given to be positive, we have $0 < x_n < r^n$ for all $n \geq N$. Since $0 < r < 1$, we know that $\lim_{n \rightarrow \infty} (r^n) = 0$. Thus, by squeeze theorem, since $\lim_{n \rightarrow \infty} (0) = \lim_{n \rightarrow \infty} (r^n) = 0$, and $0 < x_n < r^n$ for sufficiently large n , $\lim_{n \rightarrow \infty} (x_n) = 0$.

Exercise 3.2.21

- (a) Give an example of a convergent sequence (x_n) of positive numbers such that $\lim_{n \rightarrow \infty} \left((x_n)^{\frac{1}{n}} \right) = 1$.

Solution:

The sequence $x_n = 1$ trivially converges to 1 with the property that:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left((x_n)^{\frac{1}{n}} \right) &= \lim_{n \rightarrow \infty} \left((1)^{\frac{1}{n}} \right) \\
&= \lim_{n \rightarrow \infty} (1) \\
&= 1
\end{aligned}$$

- (b) Give an example of a divergent sequence (x_n) of positive numbers with $\lim_{n \rightarrow \infty} \left((x_n)^{\frac{1}{n}} \right)$. (Thus, this property cannot be used as a test for convergence).

Solution:

The sequence $(x_n) = n$ is clearly divergent (since it is unbounded), with the property that:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left((x_n)^{\frac{1}{n}} \right) &= \lim_{n \rightarrow \infty} (n)^{\frac{1}{n}} \\ &= 1\end{aligned}$$

Exercise 3.2.24 Show that if (x_n) , (y_n) , and (z_n) are convergent sequences, then the sequence (w_n) defined by $w_n := \text{mid}\{x_n, y_n, z_n\}$ is convergent.

Solution:

Solution:

Chapter 3.3: Monotone Sequences

Exercise 3.3.3 Let $x_1 \geq 2$ and $x_{n+1} := 1 + \sqrt{x_n - 1}$, $\forall n \in \mathbb{N}, n > 1$. Show that (x_n) converges and find the limit.

Solution:

To show that (x_n) is convergent, we can use the monotone convergence theorem if we are able to prove that the sequence is bounded and monotone. Let's first start with boundedness. Conjecturing that $x_n \geq 2, \forall n \in \mathbb{N}$, let $P(n)$ be the statement $P(n) : 2 \leq x_n$.

(I) Show that (x_n) is lowerbounded.

(I.i) Show the base case holds ($P(1)$ is true).

LHS:	RHS:
$= x_1$	$= 2$

Since we are given that $x_1 \geq 2$, we have $\text{LHS} \geq \text{RHS}$ and thus $P(1)$ is true.

(I.ii) Formulate induction hypothesis (State $P(k)$).

Letting $k \in \mathbb{N}$ be arbitrary, we can formulate $P(k)$ as:

$$P(k) : 2 \leq x_k$$

(I.iii) Prove $P(k) \implies P(k+1)$.

$$\begin{aligned}
 x_{k+1} &= 1 + \sqrt{x_k - 1} \\
 &\geq 1 + \sqrt{2 - 1} \\
 &= 1 + 1 \\
 &= 2
 \end{aligned}$$

Hence, $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$.

Thus, since $P(1)$ is true, and for arbitrary $k \in \mathbb{N}$, $P(k) \implies P(k+1)$, by principle of mathematical induction, $P(n)$ is true $\forall n \in \mathbb{N}$ and thus (x_n) is lowerbounded by 2.

(II) Show that (x_n) is monotonously decreasing.

Selecting $x_1 = 5$ and computing $x_2 = 1 + \sqrt{x_1 - 1} = 1 + \sqrt{4} = 3$, I conjecture that the sequence (x_n) is monotonously decreasing and thus that $x_{n+1} \leq x_n \forall n \in \mathbb{N}$. Let $P(n)$ be the statement $P(n) : x_{n+1} \leq x_n$. We can rigorously show this to be true for all choices of x_1 using induction in conjunction with the fact that $x_1 \geq 2 \implies x_1 - 1 \geq 1 \implies x_1 - 1 \geq \sqrt{x_1 - 1}$.

(II.i) Show the base case holds ($P(1)$ is true).

LHS:	RHS:
$= x_1$	$= x_2$
$= x_1$	$= 1 + \sqrt{x_1 - 1}$
$= x_1$	$\leq 1 + x_1 - 1$
$= x_1$	$= x_1$

(II.ii) Formulate induction hypothesis (State $P(k)$).

Letting $k \in \mathbb{N}$ be arbitrary, we can formulate $P(k)$ as:

$$P(k) : x_{k+1} \leq x_k$$

(II.iii) Prove $P(k) \implies P(k+1)$

$$\begin{aligned}
 x_{k+2} &= 1 + \sqrt{x_{k+1} - 1} \\
 &\leq 1 + x_{k+1} - 1 \\
 &= x_{k+1}
 \end{aligned}$$

Hence, $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$.

Thus, since $P(1)$ is true, and for arbitrary $k \in \mathbb{N}$, $P(k) \implies P(k+1)$, by principle of mathematical induction, $P(n)$ is true $\forall n \in \mathbb{N}$, and thus we know that the sequence (x_n) is monotonously decreasing.

Therefore, since (x_n) is both bounded below and monotonously decreasing by monotone convergence theorem, (x_n) converges. To find the limit, let $L = \lim_{n \rightarrow \infty} (x_n)$. Then we know:

$$\begin{aligned}
L &= \lim_{n \rightarrow \infty} (x_n) \implies \\
L &= \lim_{n \rightarrow \infty} \left(1 + \sqrt{x_{n-1} - 1} \right) \implies \\
L &= 1 + \sqrt{L - 1} \implies \\
L - 1 &= \sqrt{L - 1} \implies \\
(L - 1)^2 &= L - 1 \implies \\
L^2 - 2L + 1 &= L - 1 \implies \\
L^2 - 3L + 2 &= 0 \implies \\
(L - 2)(L - 1) &= 0
\end{aligned}$$

Thus, since we know that 2 is a lower bound of the sequence (x_n) , we know that $L = 2$ and thus $\lim_{n \rightarrow \infty} (x_n) = 2$.

Exercise 3.3.5 Let $y_1 := \sqrt{p}$ where $p > 0$ and $y_{n+1} = \sqrt{p + y_n}$, $\forall n \in \mathbb{N}$. Show that (y_n) converges and find the limit. [Hint: One upper bound is $1 + 2\sqrt{p}$].

Solution:

To prove the convergence of (y_n) , we first demonstrate that (y_n) is both upperbounded and increasing. Therefore, let $P_1(n)$ and $P_2(n)$ be the statements $P_1(n) : y_n \leq 1 + 2\sqrt{p}$ and $P_2(n) : y_n \leq y_{n+1}$ respectively. We will prove that both $P_1(n)$ and $P_2(n)$ hold $\forall n \in \mathbb{N}$ by induction.

(I) Prove (y_n) is upperbounded.

(I.i) Show the base case holds ($P_1(1)$ is true).

LHS:

$$\begin{aligned}
y_1 \\
= \sqrt{p}
\end{aligned}$$

RHS:

$$\begin{aligned}
1 + 2\sqrt{p} \\
= 1 + 2\sqrt{p}
\end{aligned}$$

Therefore, $P_1(1)$ is true.

(I.ii) Formulate induction hypothesis (State $P_1(k)$).

Letting $k \in \mathbb{N}$ be arbitrary, assume that the statement:

$$P(k) : y_k \leq 1 + 2\sqrt{p}$$

is true.

(I.iii) Prove $P_1(k) \implies P_1(k+1)$.

$$\begin{aligned} y_{k+1} &= \sqrt{p + y_k} \implies \\ (y_{k+1})^2 &= p + y_k \implies \\ (y_{k+1})^2 &\leq 1 + 2\sqrt{p} + p \implies \\ (y_{k+1})^2 &\leq (1 + \sqrt{p})^2 \implies \\ y_{k+1} &\leq 1 + \sqrt{p} \implies \\ y_{k+1} &\leq 1 + 2\sqrt{p} \end{aligned}$$

Hence, $P_1(k) \implies P_1(k+1)$ for arbitrary $k \in \mathbb{N}$.

Thus, since $P_1(1)$ is true, and for arbitrary $k \in \mathbb{N}$, $P_1(k) \implies P_1(k+1)$, by principle of mathematical induction, $P_1(n)$ is true $\forall n \in \mathbb{N}$, and thus we know that the sequence (y_n) is upper bounded by $1 + 2\sqrt{p}$.

(II) Show that (y_n) is monotonously increasing.

(II.i) Show the base case holds ($P_2(1)$ is true).

LHS	RHS:
$= y_1$	$= y_2$
$= \sqrt{p}$	$= \sqrt{p + y_1}$
$= \sqrt{p}$	$= \sqrt{p + \sqrt{p}}$
$= \sqrt{p}$	$\geq \sqrt{p}$

Thus, since $\text{LHS} \leq \text{RHS}$, $P_2(1)$ is true.

(II.ii) Formulate inductive hypothesis (State $P_2(k)$).

Letting $k \in \mathbb{N}$ be arbitrary, assume that the statement:

$$P_2(k) : y_k \leq y_{k+1}$$

is true.

(II.iii) Prove that $P_2(k) \implies P_2(k+1)$.

LHS:

$$\begin{aligned} y_{k+1} &= \sqrt{p + y_k} \\ &= \sqrt{p + y_k} \end{aligned}$$

RHS:

$$\begin{aligned} y_{k+2} &= \sqrt{p + y_{k+1}} \\ &\geq \sqrt{p + y_k} \end{aligned}$$

Hence, $P_2(k) \implies P_2(k+1)$ for arbitrary $k \in \mathbb{N}$.

Thus, since $P_2(2)$ is true, and for arbitrary $k \in \mathbb{N}$, $P_2(k) \implies P_2(k+1)$, by principle of mathematical induction, $P_2(n)$ is true $\forall n \in \mathbb{N}$, and thus we know that the sequence is monotonously increasing.

Therefore, since the sequence is upperbounded and monotonously increasing, by the monotone convergence theorem, (y_n) converges.

Let $L = \lim_{n \rightarrow \infty} (y_n)$. We therefore know that:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} (y_n) = \lim_{n \rightarrow \infty} (y_{n+1}) \implies \\ L &= \lim_{n \rightarrow \infty} (\sqrt{p + y_n}) \implies \\ L &= \sqrt{p + L} \implies \\ L^2 &= p + L \implies \\ L^2 - L - p &= 0 \implies \\ L &= \frac{1 \pm \sqrt{1 - 4(1)(p)}}{2} \implies \\ L &= \frac{1 \pm \sqrt{1 + 4p}}{2} \end{aligned}$$

For all $p > 0$, we know that $1 + 4p > 1$ and thus $\frac{1 - \sqrt{1 + 4p}}{2} < 0$ for all p . Since the sequence is increasing, and $y_1 > 0$, we know that $L > 0$ as well and thus $L = \lim_{n \rightarrow \infty} (y_n) = \frac{1 + \sqrt{1 + 4p}}{2}$.

Exercise 3.3.8 Let (a_n) be an increasing sequence, (b_n) be a decreasing sequence, and assume that $a_n \leq b_n$, $\forall n \in \mathbb{N}$. Show that $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$, and thereby deduce the Nested Interval Property from the Monotone Convergence Theorem.

Solution:

(I) Prove $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$.

Proof. We know that (b_n) is a decreasing sequence and lower bounded by any term (such as a_1) in a_n . Therefore, by the Monotone Convergence Theorem, we know that $\lim_{n \rightarrow \infty} (b_n)$ exists. Likewise, we know $\lim_{n \rightarrow \infty} (a_n)$ exists, since (a_n) is an increasing sequence and bounded above by any term in (b_n) . Thus, since (a_n) and (b_n) are both convergent sequences satisfying $a_n \leq b_n$ $\forall n \in \mathbb{N}$, we can invoke Theorem 3.2.5 to conclude that $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$. $\square$

(II) Prove the Nested Interval Property using the Monotone Convergence Theorem.

Proof. Let $I_1 \supseteq I_2 \supseteq \dots \supseteq I_n \supseteq \dots$ be a sequence of non-empty, nested, and closed intervals where $I_n = [a_n, b_n]$. We know, for the sequences to be nested, that $a_n \leq a_{n+1}$ and $b_n \geq b_{n+1}$, $\forall n \in \mathbb{N}$ (see Exercise 2.5.6 for proof), implying that the sequence (a_n) and (b_n) are increasing and decreasing sequences respectively. Furthermore, for each I_n to be non-empty, we also know that $a_n \leq b_n$. Thus, using our prior findings, we know $\lim_{n \rightarrow \infty} (a_n)$ and $\lim_{n \rightarrow \infty} (b_n)$ both exist with $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$. Now consider $\bigcap_{n=0}^{\infty} I_n$. Since each set is nested, we know that $\bigcap_{n=0}^{\infty} I_n = [\lim_{n \rightarrow \infty} (a_n), \lim_{n \rightarrow \infty} (b_n)]$. However, since $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$, we know that the closed interval $[\lim_{n \rightarrow \infty} (a_n), \lim_{n \rightarrow \infty} (b_n)]$ is non-empty and hence we have derived the Nested Interval Property from the Monotone Convergence Theorem. $\square$

Exercise 3.3.9 Let A be an infinite subset of $\mathbb{R}$ that is bounded above and let $u := \sup(A)$. Show that there exists an increasing sequence (x_n) with $x_n \in A$ $\forall n \in \mathbb{N}$ such that $\lim_{n \rightarrow \infty} (x_n) = u$.

Solution:

Proof. If $u \in A$, then trivially we know that the sequence $(x_n) = u$ is an increasing sequence (since $x_n \leq x_{n+1}$, $\forall n \in \mathbb{N}$) that converges to u . If $u \notin A$, then we can rely on the definition of the supremum to construct x_n . Consider the sequence $\epsilon_n = \frac{1}{n}$. By definition of supremum, we know that $\exists a_n, a_{n+1} \in A$ such that $u - \epsilon_n < a_n$ and $u - \epsilon_{n+1} < a_{n+1}$. Using this notion, we can

construct an increasing (x_n) such that $x_1 \in (a_1, a_2)$, $x_2 \in (a_2, a_3)$, $\dots$, $x_n \in (a_n, a_{n+1})$, $\dots$. To prove that (x_n) converges to u , consider any $\epsilon > 0$. By the Archimedean principle, we know that there exists an $N \in \mathbb{N}$ such that $\epsilon_N = \frac{1}{N} < \epsilon$. This also clearly means that $\forall n \geq N, \epsilon_n = \frac{1}{n} < \epsilon$. Thus, $\forall x_n, n \geq N$, we have that:

$$\begin{aligned} x_n \in (a_n, a_{n+1}) &\implies \\ a_n < x_n < a_{n+1} &\implies \\ u - \epsilon_n < x_n &\implies \\ u - x_n < \epsilon_n &\implies \\ u - x_n < \epsilon &\implies \\ |u - x_n| < \epsilon \end{aligned}$$

Therefore, by definition of convergence, the sequence (x_n) converges and therefore, for all infinite subsets of $\mathbb{R}$ which are bounded above, there always exists a sequence contained within the set which converges to its supremum. $\square$

Exercise 3.3.10 Establish the convergence or divergence of the sequence $(y_n) := \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$, $\forall n \in \mathbb{N}$.

Solution:

I conjecture that (y_n) is a convergent sequence and will attempt to prove this using the monotone convergence theorem by showing that (y_n) is both upperbounded and increasing.

(I) Prove (y_n) is bounded above.

$$\begin{aligned} y_n &= \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \\ &\leq \frac{1}{n+1} + \frac{1}{n+1} + \dots + \frac{1}{n+1} \\ &= \frac{n}{n+1} \\ &= 1 - \frac{1}{n+1} \\ &< 1 \end{aligned}$$

Thus, the sequence (y_n) is bounded above by 1.

(II) Prove (y_n) is monotonously increasing.

We can likewise prove that (y_n) is increasing simply through algebraic manipulation

$$\begin{aligned}
 y_{n+1} &= \frac{1}{(n+1)+1} + \frac{1}{(n+1)+2} + \dots + \frac{1}{(n+1)+n-1} + \frac{1}{(n+1)+n} + \frac{1}{(n+1)+n+1} \\
 &= \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} + \frac{1}{2n+1} + \frac{1}{2n+2} \\
 &= \frac{1}{n+1} - \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} + \frac{1}{2n+1} + \frac{1}{2n+2} \\
 &= \left(\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} \right) + \frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1} \\
 &= y_n + \frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1} \\
 &\geq y_n + \frac{1}{2n+2} + \frac{1}{2n+2} - \frac{1}{n+1} \\
 &= y_n + \frac{1+1-2}{2n+2} \\
 &= y_n + 0 \\
 &= y_n
 \end{aligned}$$

Therefore, since $y_{n+1} \geq y_n$, $\forall n \in \mathbb{N}$, the sequence (y_n) is increasing.

Thus, since (y_n) is bounded above and increasing, by the Monotone Convergence Theorem, (y_n) converges.

Exercise 3.3.12 Establish the convergence and find the limit of the following sequences:

(a) $\left(1 + \frac{1}{n}\right)^{n+1}$.

Solution: